NutriMatch

System review and Q&A

A study guide to the whole system: what it does, how each part works, the rules behind it, and the questions a panel is likely to ask.

NutriMatch: Web-Based Clinical Nutrition Consultation Management System
Capstone project, STI College Davao, BSIT
Reflects the system as of November 2, 2026 (branch rnd-design-pass)

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> How to use this guide: read the "How it works" part of each chapter first, then cover the answers and try the Q&A on your own. The panel questions in chapter 20 are the ones most worth practising out loud.

1. The system at a glance

NutriMatch connects clients with licensed Registered Nutritionist-Dietitians (RNDs) in the Philippines for Medical Nutrition Therapy. Care follows the clinical **Nutrition Care Process** (Assessment, Diagnosis, Intervention, Monitoring and Evaluation) of the Academy of Nutrition and Dietetics. Around that clinical core the system handles booking, payments, video consultations, meal plans, lab results, progress tracking, reminders and administration.

Part	What is used
Frontend	Nuxt 4 (Vue), Tailwind CSS, Lucide icons. Hosted on Vercel.
Backend	Django 6 with Django REST Framework, SimpleJWT sign-in, Django Channels for live features. Hosted on Render.
Database	PostgreSQL on Neon (SQLite during development). About 40 tables.
Files	Cloudinary. PRC licenses, lab results and meal photos are private.
Payments	PayMongo Checkout (GCash, Maya, cards, QR Ph), test mode.
Video	Our own browser-to-browser WebRTC call, with a Cloudflare relay (TURN).
Email and SMS	Brevo (email in production), Semaphore (SMS).
Food data	FNRI Food Exchange Lists (main), USDA FoodData Central (second source).

Live addresses: frontend `nutrimatch-theta.vercel.app`, API `nutrimatch-api.onrender.com/api`.

Q. In one sentence, what problem does NutriMatch solve?

It gives Filipino clients structured, documented nutrition care from licensed RNDs online, and gives RNDs one place to run that care from screening to meal plan to follow-up.

Q. Who are the three users?

Clients (patients), RNDs (licensed dietitians, approved by an admin) and admins (NutriMatch staff who verify RNDs, handle support, refunds, payouts and reports).

Q. Is the NCP something the team invented?

No. It is the standard clinical method of the Academy of Nutrition and Dietetics. The system follows its four phases and its standard terminology (PES statements, etiology matrix, assessment terms).

2. Roles and their journeys

Client

1. Signs up, confirms the 6-digit email code, accepts the Terms and gives health-data consent.
2. Onboarding: date of birth and sex, health background (allergies are required), then the pre-consultation screening.
3. Finds an RND (filters by specialty, language, format, fee) and books a time.

4. Pays after the RND confirms, at least 15 minutes before the start.
5. Joins the video call (or meets at the clinic for in-person).
6. Receives a consultation summary, a meal plan and lab requests; logs meals with a photo or note.
7. Follows progress, books follow-ups, and can request a discharge.

RND

1. Applies with PRC number, license photo, specialization, fee and formats; an admin approves.
2. Sets weekly hours (with breaks), time off, bio and payout details.
3. Confirms or declines booking requests (with the client's screening summary).
4. Runs the consultation: video, patient panel, NCP form, lab requests.
5. Finalizes the NCP record, builds and sends the weekly meal plan, rates meal logs.
6. Sees earnings; gets paid monthly; discharges the patient when care is done.

Admin

1. Reviews RND applications (approve or reject with a reason).
2. Watches the dashboard's Needs attention list.
3. Answers support requests and decides no-show disputes.
4. Sends refunds through PayMongo and releases monthly RND payouts.
5. Reads reports and the audit log; manages users and admins.

3. Accounts, sign-up and consent

How it works

- **Email verification:** a 6-digit code by email; sign-in is refused until it is confirmed.
- **Passwords:** upper case, lower case, a number and a special character. Stored hashed (PBKDF2), never readable, even by admins.
- **Ages:** clients 18 or older, RNDs 21 or older.
- **Consent (RA 10173):** two separate boxes the person ticks themselves: the Terms and Privacy Policy, and consent to process health data. Both stay locked until the person opens both documents. The time of consent is saved and logged.
- **RND application:** 7-digit PRC number, a photo of the license (stored privately), specialization, fee (at least ₱100), formats (video, in person), optional bio. An admin approves or rejects with a reason; the RND is emailed.
- **Mobile numbers:** optional for clients (any country), required and Philippine-only for RNDs.
- **An RND is visible to clients** only when approved, with a fee, a bio and at least one block of weekly hours.

Q. Why are there two consent boxes instead of one?

The Data Privacy Act treats health data as sensitive personal information. Consent to process it must be specific, so it is kept separate from accepting the Terms, and nothing ticks either box automatically.

Q. How is an RND's license checked?

The RND uploads a photo of the PRC license and types the license number. An admin compares them on the RND Verification page and approves or rejects. Until approved, the RND cannot sign in to any RND feature.

Q. What if someone signs up but never confirms the email?

Registering again with the same email replaces the unfinished account. Only a verified account blocks the email.

4. Finding an RND and booking

- **Find an RND** shows only bookable RNDs, soonest available first. Filters (specialty, language, format, maximum fee) only list values some RND actually has.
- **Client-facing RND data is public information only:** name, photo, specialization, bio, fee, formats, languages, PRC number, rating. Never email or phone.
- **One RND at a time:** a client with active care cannot book a different RND until discharged.
- **Booking times** start every hour from the start of the RND's block (a block 8:00 AM to 12:00 PM gives 8, 9, 10, 11 AM). Booked and blocked times stay visible but greyed out.
- **A first booking needs a screening** before the RND can confirm it.
- **Availability:** several blocks a day are allowed (for example a lunch break: 8 to 12 and 1 to 5), never overlapping. A block can end at 12:00 AM, meaning midnight. Each day shows a "Clients can book..." preview. Time off blocks whole days or part of a day.

Q. Why only one RND at a time?

Clinical safety and clear responsibility: two RNDs could give conflicting plans. A client who wants to change can request a discharge, which ends automatically after 3 days if the RND does nothing, so nobody is trapped.

Q. How does an RND set a lunch break?

By adding two blocks on the same day, for example 8:00 AM to 12:00 PM and 1:00 PM to 5:00 PM. The page shows exactly which hours clients can book, so the gap is visible.

5. Time rules for consultations

These rules protect both sides: the client always has time to pay, and the RND knows early whether a session is happening.

Example: consultation at 3:00 PM	Rule
Latest booking 2:15 PM	A booking must start at least 45 minutes from now.
RND confirms by 2:30 PM	Otherwise the request expires and the client is told.
Paid client can cancel until 2:30 PM	Full refund. After that, no cancelling.
Client pays by 2:45 PM	Any time after confirming. Still unpaid at 2:45 the booking is cancelled automatically and the invoice voided.
Room opens 2:45 PM	15 minutes before the start; closes 15 minutes after the scheduled end.

Q. Can a client pay a day early?

Yes. They can pay any time after the RND confirms. 15 minutes before the start is only the last moment. The texts say "Pay any time before Nov 3, 2:45 PM" so it is not mistaken for today.

Q. Why 45 minutes ahead?

It is 30 minutes for the RND to confirm plus 15 minutes for the client to pay. Anything shorter could leave the client no time to pay.

Q. What if the RND never answers a request?

It expires 30 minutes before the start. Both are notified and the client can book another time.

6. Payments, refunds and payouts

Paying

- The invoice is created when the RND confirms. The platform commission (default 10%) is **frozen** on the invoice at that moment.
- The client pays through a PayMongo Checkout page: GCash, Maya, card or QR Ph. PayMongo tells the server through a signed webhook; the Billing page also checks PayMongo directly in case a webhook is missed.
- Every payment sends a branded email receipt. Receipts are payment receipts, not BIR official receipts.
- The client only gets into the video call once paid.

Refunds

- Always the full amount, within 5 to 7 business days, started by an admin through PayMongo.
- GCash, Maya and card payments go back to the same account. QR Ph payments get a PayMongo claim link.
- An invoice only turns "Refunded" when PayMongo reports the refund succeeded.

RND payouts

- RND share = fee minus the frozen commission.
- On hold for 3 days after the consultation is completed (or marked a no-show), then cleared.
- Paid monthly on the 5th, only balances of at least ₱500 that cleared by that day; smaller balances carry over. No early payouts.

- Money is sent outside the app (GCash or bank); the admin records the receipt reference. Safeguards: the amount must match, the reference must be real, and changed payout details within 7 days need extra confirmation.

Q. Why is the commission frozen?

So changing the commission rate later never changes what was already agreed on past invoices.

Q. Is real money involved?

No. PayMongo runs in test mode. Going live needs PayMongo business verification. Note that the PayMongo sandbox cannot complete QR Ph refunds; GCash, Maya and card refunds work in test mode.

7. The consultation room and video calls

- **Video is our own WebRTC call:** browser to browser, encrypted by WebRTC, one-on-one, never recorded. A Cloudflare relay (TURN) helps calls connect on mobile data and strict networks; it relays the encrypted stream without storing it. The server only introduces the two browsers (signaling) and checks who may join.
- **Who may join:** the appointment's RND and client, a confirmed video consultation, inside the window (15 minutes before to 15 minutes after), and the client must have paid.
- **RND side panel:** Patient (allergies, figures, care plan, meal logs, labs), NCP (the record form) and lab requests.
- **Attendance:** the server records when each person joined and left and the minutes in the call, only times, nothing from the call.
- **Rejoining:** both can leave and rejoin until the RND finishes or the room closes. A busy camera is retried, the client rejoins unmuted, and a "Tap to hear" button appears if the browser blocks sound.
- **Finish consultation (RND)** has four choices: It's done (mark complete), the patient didn't show up, the video call didn't work, leave for now.

Q. Why not Jitsi or Zoom?

Embedded Jitsi needed a moderator login and cut calls after about 5 minutes; Daily.co needed a credit card. Our own WebRTC call has no login, no time limit, no cost besides the free Cloudflare relay tier, and keeps calls private.

Q. Are calls recorded?

No. Only join and leave times are stored, for attendance and disputes. This follows RA 10173's data minimization.

Q. Why do two laptops in the same room howl?

One device's microphone picks up the other's speakers. Echo cancellation can only remove a device's own sound. Use earphones or separate rooms; it is not a bug.

8. No-shows, disputes and cancellations

Situation	What happens	Money
Client never joins	RND marks a no-show (only after the start and only if the client never joined the call)	No refund; RND is paid. Client may dispute within 3 days.
Client disputes	Opens a support request with the reason; admin sees the attendance record and decides	Payout waits. Admin approves (full refund) or rejects (RND keeps it).
RND doesn't show	15 minutes after the start, if the RND was in the call under 5 minutes in total, the client reports it	Full refund; counts against the RND.
Call doesn't work	RND ends it as a failed call	Full refund; RND not paid; not counted against the RND.
RND emergency (last 30 minutes)	RND picks a reason: emergency, technical or other	Full refund. 3 late cancels in 30 days flags the RND for the admin.
Client cancels	Paid: up to 30 minutes before. Unpaid: any time before the pay deadline	Paid: full refund.

Q. Why have a Dispute button if there is already Help & Support?

A dispute is a support request underneath, but the button also holds the RND's payout, attaches the attendance record, gives the admin Approve/Reject buttons that move the money, and enforces the 3-day deadline.

Q. Can an RND fake a no-show?

Not for a video call the client joined: the server refuses it. For in-person consultations there is no call record, so the client's dispute and the admin's judgement are the check.

Q. Why can a brief RND join still count as not attending?

The rule uses total minutes in the call (under 5 minutes), not "ever joined", so joining for a few seconds doesn't block the client's report.

9. Screening and the calculations

The pre-consultation screening is taken once at onboarding and then locked. It is the starting point that progress is measured from; newer measurements come from the RND's NCP records.

Formulas

Value	Formula
BMI	weight (kg) ÷ height (m)^2. Classified with the **WHO Asia-Pacific** cut-offs: under 18.5 underweight, 18.5 to 22.9 normal, 23 to 24.9 overweight, 25 to 29.9 Obese I, 30 and over Obese II.
Desirable body weight (DBW)	**BMI method:** height (m)^2 × 22.
BMR	DBW × 1.0 (male) or 0.9 (female) kcal per kg per hour × 24.
TER (Tannhauser factorial)	BMR + physical activity (bed rest 10%, sedentary 30%, light 50%, moderate 75%, heavy 100% of BMR) + TEF (10% of BMR plus activity), rounded to the nearest 50 kcal.
Macros	TER × % ÷ 4 kcal/g (carbohydrate, protein) or ÷ 9 kcal/g (fat), rounded to 5 g. Default 55 / 20 / 25%.
NRS-2002	Nutritional status (BMI under 18.5 = 3; weight loss of 5% or more since the last screening = 2; unintended weight loss or poor appetite = 1) plus disease severity (1 for elevated glucose or recent hospitalization). 3 or more = at risk, 5 or more = high risk.

> **Worked example** (male, 168 cm, light activity): DBW = 1.68^2 × 22 = 62.09 kg. BMR = 62.09 × 1.0 × 24 = 1,490 kcal. Activity = 50% = 745 kcal. TEF = 10% of 2,235 = 224 kcal. Total 2,459, so **TER = 2,450 kcal**. Macros at 55/20/25: carbohydrate 335 g, protein 125 g, fat 70 g.

Q. The proposal says Mifflin-St Jeor. Why is the BMR different?

On the request of our RND resource person we switched to Tannhauser's desirable-weight method, which is how Philippine dietetics computes energy needs. Tannhauser's DBW, (height − 100) − 10%, gives 0 kg at 100 cm and fails for short stature, so the team adopted the BMI method (height^2 × 22) for the DBW. The RND confirmed the BMI method for short clients.

Q. Why WHO Asia-Pacific BMI cut-offs?

Health risks for Asian adults start at lower BMI values, so the Asia-Pacific ranges are lower than the global ones (for example, overweight starts at 23, not 25).

Q. What about very short clients?

Below 150 cm the BMR and BMI carry a note that the estimate may be less accurate and the RND reviews it. Heights from 100 cm are allowed. The RND always sets the final target.

Q. Why is the screening locked?

It is the baseline. If a client could edit it, weight change and BMI change could not be measured honestly. New measurements come from the RND's NCP record, and the client's screening page shows those newer numbers too.

10. The Nutrition Care Process (NCP) record

- **One record, four phases:** Assessment, Diagnosis, Intervention, then Monitoring on follow-ups. Stored as one longitudinal row, not four tables.

- **Started in the consultation room** and linked to that consultation; continued later from the patient chart.
- **Assessment:** height, weight, blood pressure, fasting glucose, HbA1c (pre-filled from the latest records and locked until "Edit measurements"; a changed value shows what it was, with Put back), other measurements, lab and assessment notes. Warning labels for BP, glucose and HbA1c categories (never blocking).
- **Diagnosis (PES):** Problem, Etiology and Signs/symptoms, each a searchable picker from the AND's official lists (238 etiologies from the 2020 Etiology Matrix; 1,147 assessment terms), free text still allowed.
- **Intervention:** calculated TER with "Show calculation", the TER the RND sets, macros in %, grams calculated, other targets, and the diet prescription (its first lines are written from the numbers).
- **Monitoring** (follow-ups): previous goals, weight change since, adherence, and goal status.
- **Save draft** any time; **Finalize** needs the clinical minimum (height, weight, PES, prescription, TER, goal status on follow-ups). A finalized record is permanent and no NCP record can be deleted.
- Finalizing copies the measurements into the client's Progress Tracker.

Q. Why can't an NCP record be deleted, even a draft?
A record only starts from a real consultation, so even a draft is clinical work. Keeping it preserves an honest medical history.

Q. Where do the dropdown terms come from?
From the Academy of Nutrition and Dietetics' 2020 Etiology Matrix and the 2020 Nutrition Assessment and Monitoring terminology sheet (both in the vault folder). The wording is copied, not invented.

11. Lab results

- The RND requests tests from the chart or the room; "needed before" is always the next consultation.
- The client uploads PDF, JPG or PNG files (up to 5 files, 10 MB each) with the test date and, for glucose, whether they fasted.
- Files are private Cloudinary files streamed only through the app; admins cannot open them. Opening one is logged.
- The RND accepts (final) or declines with a reason (the client uploads again).
- Values are typed into the NCP by the RND; nothing is read from the files automatically.
- Discharging a client cancels any lab request still open.

Q. Why can't admins see lab files?
They are health records needed only by the treating RND. Data minimization: only the people who need them can open them.

12. Meal planning and meal logs

- **Weekly plans**, meals per day with times, built from FNRI foods (in exchanges) or USDA foods (in grams).
- **FNRI exchanges:** the RND sets the number of exchanges (in halves); calories and macros follow, and the portion reads like "2 × ½ cup".
- **Diet templates:** diabetes, hypertension (DASH), renal, weight management, weight gain, general. They start from the NCP target (else the screening TER), fill a full week, use foods allowed for the condition, and skip the client's allergens. Weight management cuts 500 kcal or 20% of the TER, whichever is smaller.

- **Warnings:** foods that may not suit the condition are amber; allergens are red. Sending asks for confirmation, and if an allergen is in the plan the RND must tick a box confirming it is safe.
- **Draft vs sent:** drafts can be deleted; a sent plan cannot (the client logs against it). Undo and Discard changes are available while editing.
- **Meal logs:** the client logs a photo and/or note; the server stamps the time. The RND rates it Followed, Partly or Not followed, which is final.
- Meal photos are private and their links expire after 10 minutes.
- The client can **Download / print** the whole week's plan.

Q. Why store only the food name from USDA or FNRI?

Data minimization from the proposal: the system keeps only what it needs (the food name and the amount), not whole external records.

Q. How is adherence calculated?

Followed counts 1, partly followed 0.5, not followed or not logged 0, over the planned meals since the plan was sent. Logs still waiting for a rating are left out.

13. Progress, adherence and care stages

- **Progress Tracker (client):** weight, BMI, blood pressure, glucose and HbA1c charts from the screening and the RND's finalized NCP records, weekly adherence, and a printable report.
- **Patient chart (RND):** Overview, NCP records timeline, Meal plan, Progress (trend charts), Lab results, Consultations, Tasks and reminders.
- **Care stages:** screening, NCP, meal plan, follow-up, with a "next step" for the RND.
- **Alerts** on the RND dashboard: no logs for 3 days, adherence under 50%, NRS-2002 risk, a big weight change, an idle NCP draft.
- **Consultation summary** after each completed visit, printable; the client never sees the RND's private notes.

14. Messages, notifications, email and SMS

- **Messages:** client-RND chat, live, only while care is active (history stays readable).
- **Notifications:** in-app, live; a bell for clients and RNDs.
- **Reminders:** 24 hours and 1 hour before a consultation, payment due, meal logging, lab uploads, and tasks the RND sets (the client ticks them off in My tasks).
- **Emails always sent:** booking confirmed with the pay-by time, cancelled by the RND, marked as not attending, receipts, approvals, care ended, support replies. Reminder emails are opt-in.
- **SMS (Semaphore):** opt-in, Philippine mobile only. Four texts: the 1-hour reminder, payment due, booking confirmed, cancellation. Texts never contain health details. Waiting for Semaphore to approve the "NutriMatch" sender name.

Q. Why no health details in texts?

An SMS can be read by anyone holding the phone. Health information stays inside the app, where access is controlled.

15. Discharge and ending care

- The RND discharges with a reason. It is blocked while a consultation is upcoming or not yet completed, or while an NCP draft is open.
- Discharge archives the active meal plan, stops the RND's reminders, cancels open lab requests, logs it, and notifies and emails the client.
- The client can request a discharge (with a reason); if the RND does nothing within 3 days, care ends automatically.
- A discharged record is read-only. The client keeps their history and can book the same or another RND; booking the same RND again reopens the care.

16. The admin side

Page	What it is for
Dashboard	Today's figures, Needs attention (unpaid soon, waiting requests, support, applications, refunds, payouts due, disputes, RNDs with many late cancels), month vs last month.
Users	Every account; deactivate or reactivate with a reason; send a password reset; edit details with a reason; add admins.
RND Verification	Review applications: license photo and number; approve or reject.
Appointments	Every consultation with its payment and attendance label (green both attended, red only one or neither); cancel with a reason.
Support	Requests from clients and RNDs; reply, close; decide no-show disputes.
Billing and commission	Invoices, receipts, refunds through PayMongo.
Payouts	Monthly payout runs with safeguards.
Reports	Consultations, payments, commission, new clients, per RND; print or CSV.
Audit log	Who did what and when (immutable).

Q. Can an admin see a client's health record?

Yes, for support and compliance, and every view is logged in the audit log. Admins cannot open lab files or meal photos, and never see passwords.

17. Privacy and security (RA 10173)

- **Consent:** separate, explicit consent for health data, with timestamps.
- **Access control:** role-based; an RND only sees their own patients; clinical data only within active care.
- **Encryption in transit:** HTTPS everywhere; video encrypted by WebRTC.
- **Passwords:** salted PBKDF2 hashes.
- **Private files:** licenses, lab results and meal photos are private, opened only through the app, photo links expire in 10 minutes.

- **Audit log:** immutable rows for significant actions (consent, record views, refunds, discharges, edits).
- **Data minimization:** only food names from external sources; no call recordings; no health data in SMS.
- **Out of scope (per the proposal):** NPC registration, end-to-end encryption, automatic retention schedules, granular consent withdrawal.

Q. Is NutriMatch registered with the National Privacy Commission?

No. It is an academic capstone. Anyone deploying it for real clinical use would have to register it; the Privacy Policy says so.

18. Architecture and deployment

Backend apps (Django)

- **accounts:** users, sign-in, verification, admin user tools
- **profiles:** RND and client profiles, availability, time off
- **scheduling:** relationships, appointments, video call, attendance, no-shows, discharge
- **clinical:** screening, NCP, progress, labs, calculations
- **nutrition:** FNRI foods, meal plans, templates, meal logs, USDA
- **communication:** messages, notifications, reminders, SMS, support
- **billing:** invoices, PayMongo, refunds, payouts
- **core:** settings, audit log, reports, admin dashboard

How a request flows

1. The Nuxt page calls the REST API with a JWT.
2. Django checks the role and ownership, applies the rules, saves to PostgreSQL.
3. Side effects run after the save: notifications (pushed live over WebSockets), emails, SMS, audit rows.
4. External services: PayMongo, Cloudinary, Brevo, Semaphore, Cloudflare TURN, USDA.

- **Deployment:** pushing to the main branch deploys automatically: Render (backend, migrations run on start) and Vercel (frontend). Free tiers; Render sleeps after 15 minutes idle (about a minute to wake).
- **Time:** all "today" decisions use Philippine time on the server's clock, so changing a phone's date changes nothing.
- **Testing:** about 570 automated backend tests run before every push.

19. Changes from the proposal

Proposal	Built system	Why
Laravel 11, MySQL	Django, PostgreSQL	Team decision; Django's admin, ORM and REST framework fit; free hosting on Render and Neon.
21 tables	About 40 tables	Added features: labs, time off, support, payouts, resource sharing, attendance and more.
Mifflin-St Jeor BMR, TDEE	BMI-method DBW, Tannhauser TER	RND resource person's method; the BMI method handles short stature.
Chat consultations	Video or in person only	Adviser's call; chat stays as messaging.
Third-party video (Jitsi)	Own WebRTC call	No login, no time limit, private.
Payment after the consultation	Pay before the consultation	Protects the RND's time; clear refund rules.
bcrypt	PBKDF2	Django's standard password hashing.

20. Likely panel questions

Q. Walk us through what happens from booking to meal plan.

The client books a time at least 45 minutes ahead; the RND reviews the screening and confirms; an invoice is created and the client pays; at the time they meet in the video room; the RND fills the NCP record, finalizes it, generates a meal plan from a template, checks it and sends it; the client logs meals and the RND rates them; progress shows on both sides.

Q. How do you make sure the calculations are correct?

They use named clinical formulas (WHO Asia-Pacific BMI, BMI-method DBW, Tannhauser TER, NRS-2002), were checked against hand calculations, and are covered by automated tests. The steps are shown to the RND with "Show calculation".

Q. How do you protect patients with allergies?

Allergies are required at onboarding. Templates skip allergens. Allergens show in red in every food picker and plan, and sending a plan with an allergen requires the RND to confirm it is safe.

Q. What stops a client from not paying and still joining?

The server only lets a paid client into the call, and unpaid bookings are cancelled 15 minutes before the start.

Q. What if the RND's internet fails during the call?

The call reconnects by itself; both can rejoin. If it can't work at all, the RND ends it as a failed call and the client is refunded in full.

Q. How is data privacy handled?

Explicit separate consent, role-based access, HTTPS, hashed passwords, private files with expiring links, no call recordings, no health details in SMS, and an immutable audit log, including every admin view of a client record.

Q. Why should a client trust an RND on NutriMatch?

Every RND's PRC license is checked by an admin before they can practice on the platform, the PRC number is shown on their profile, and clients leave ratings and reviews.

Q. How does NutriMatch earn?

A platform commission (default 10%) on each paid consultation, frozen on the invoice. RNDs receive the rest monthly.

Q. Can it scale?

The backend is stateless apart from live features; with more traffic, Render can run more instances with Redis for live updates (supported by the code via REDIS_URL), and Neon PostgreSQL scales separately.

Q. What would you add next?

Live PayMongo after business verification, SMS once the sender name is approved, automatic payouts, a client data download, and validated energy equations for special populations.

21. Known limitations and open items

- PayMongo is in test mode; QR Ph refunds can't complete in the sandbox.
- SMS waits for Semaphore to approve the "NutriMatch" sender name.
- The minimum calorie target is 800 kcal; very short clients may have a TER below it (waiting on the RND resource person).
- BMR and TER are estimates from desirable weight; the RND sets the final target.
- Payouts are recorded manually.
- In-person no-shows have no attendance record; disputes rely on the admin's judgement.
- The free Render plan sleeps when idle (about a minute to wake).
- Not NPC-registered (academic project).

22. Glossary

RND	Registered Nutritionist-Dietitian, licensed by the PRC.
PRC	Professional Regulation Commission.
MNT	Medical Nutrition Therapy.
NCP	Nutrition Care Process: Assessment, Diagnosis, Intervention, Monitoring and Evaluation.
PES	Problem, Etiology, Signs/symptoms: the format of a nutrition diagnosis.
DBW	Desirable body weight.
BMR	Basal metabolic rate: calories burned at rest.
TER	Total Energy Requirement: calories needed per day.
TEF	Thermic effect of food (energy to digest food, 10%).
NRS-2002	Nutritional Risk Screening 2002.
FNRI	Food and Nutrition Research Institute; its Food Exchange Lists.
Exchange	A standard portion of a food group with known calories and macros.
WebRTC	Browser technology for direct, encrypted audio and video.
TURN	A relay server that helps calls connect through strict networks.
RA 10173	Data Privacy Act of 2012.
JWT	JSON Web Token, used for signing in to the API.